

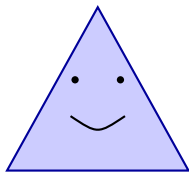
The Banach Mazur Game

A Dynamic View of Topological Largeness

Eli Griffiths

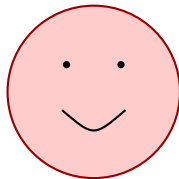
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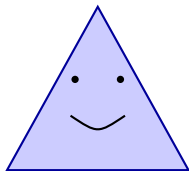


Alice

vs.

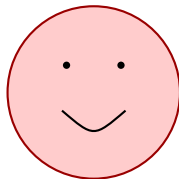


Bob



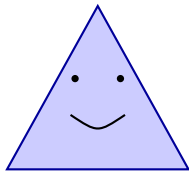
Alice

vs.



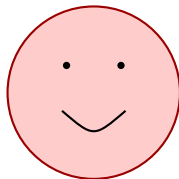
Bob

Let $A \subset \mathbb{R}$ be given.



Alice

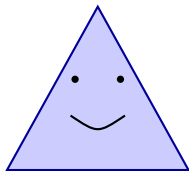
vs.



Bob

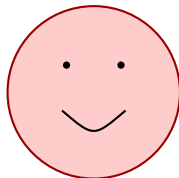
A *run* of the game is a sequence of open non-empty intervals U_n and V_n such that

$$U_1 \supset V_1 \supset U_2 \supset V_2 \supset \dots$$



Alice

vs.



Bob

Bob wins if

$$A \cap (U_1 \cap V_1 \cap U_2 \cap V_2 \dots) = \emptyset.$$

Otherwise Alice wins.

What choices of A can Bob always win?

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- A is finite

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- A is finite
- A is countably infinite

Definition (Strategy)

A **strategy** for Bob is a sequence of maps

$$\sigma_n : (U_0, V_0, \dots, U_n) \mapsto V_n \subset U_n.$$

A strategy may also be of the form $\sigma_n(U_n) = V_n$, relying on just the last move.

Definition (Winning Strategy)

A strategy $(\sigma_n)_{n \in \mathbb{N}}$ for Bob is winning if for any run

$$U_1 \supset V_1 \supset U_2 \supset V_2 \supset \dots$$

where $\sigma_n(U_1, V_1, \dots, U_n) = V_n$, it holds that

$$A \cap \bigcap_{n \in \mathbb{N}} U_n = A \cap \bigcap_{n \in \mathbb{N}} V_n = \emptyset.$$

Theorem (Meager $\Rightarrow$ Strategy)

If A is meager, then Bob has a winning strategy.

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Theorem (Strategy $\Rightarrow$ Meager)

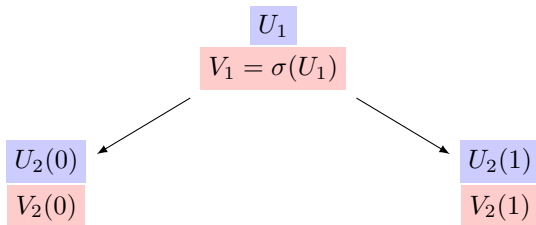
If Bob has a winning strategy, then A is meager

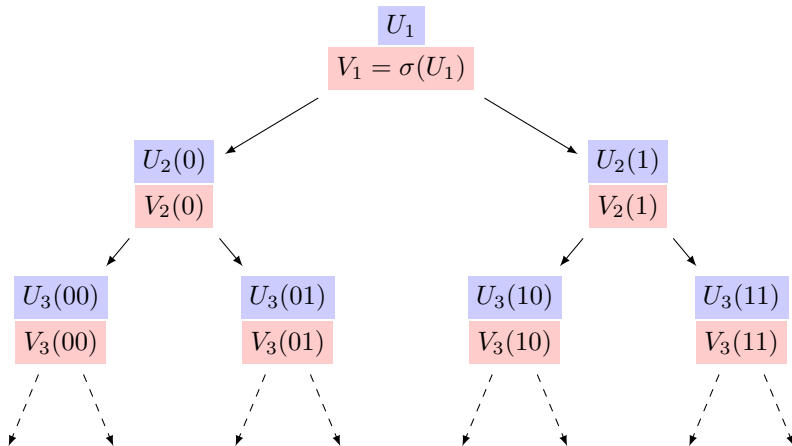
Theorem

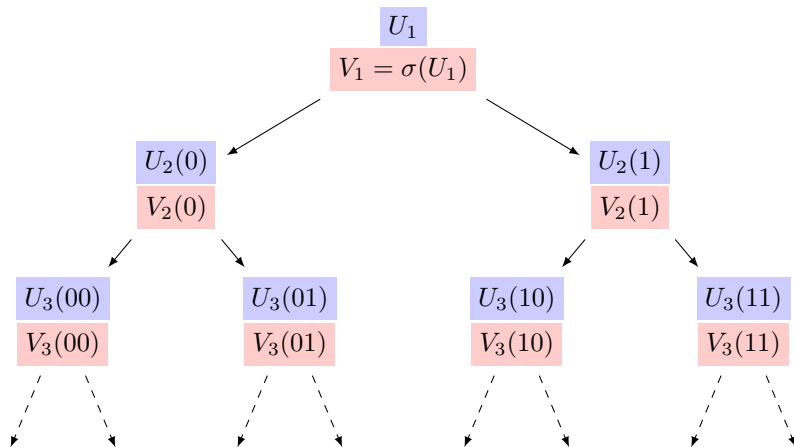
If Bob has a winning strategy, then $|A^c| \geq |\mathbb{R}|$.

$$U_1$$

$$V_1 = \sigma(U_1)$$







$$b = (b_1, b_2, \dots) \in \{0, 1\}^{\mathbb{N}} \quad \rightsquigarrow \quad (U_n(b_1, \dots, b_n), V_n(b_1, \dots, b_n))_{n \geq 1}$$

Let X be a topological space, with $A \subset X$ and Alice and Bob's set choices being non-empty open sets in X . Winning conditions are the same.

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Definition (Completely Metrizable)

X is completely metrizable if there exists a complete metric d on X that induces the topology of X ; that is, the set of all open d -balls generates X .

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Definition (Completely Metrizable)

X is completely metrizable if there exists a complete metric d on X that induces the topology of X ; that is, the set of all open d -balls generates X .

Theorem

Let X be completely metrizable. If Bob has a winning strategy, then A is meager.

Definition (Modified Banach Mazur)

Alice and Bob similarly choose $U_1 \supset V_1 \supset \dots$ such that $U_n, V_n \in \tau_X$. Bob wins if

$$\bigcap_{n \in \mathbb{N}} U_n = \bigcap_{n \in \mathbb{N}} V_n = \emptyset.$$

Otherwise Alice wins. Denote this modified game as $BM(X)$.

Definition (Baire Space)

A space X is Baire if for any sequence U_n of dense open sets, $\bigcap U_n$ is dense.

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Theorem

A space X is Baire if and only if Alice has no winning strategy in $BM(X)$.

1. Intuition for Topological Size and Baire Spaces

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2. Utility in Proving Results (Ex. 1 and Ex. 2)

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3. Historical Relevancy